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Wifaro

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```r
library(survival)
library(survminer)
```

```
## Warning: package 'survminer' was built under R version 4.5.2

## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.5.2

## Loading required package: ggpubr

## Warning: package 'ggpubr' was built under R version 4.5.2

##
## Attaching package: 'survminer'

## The following object is masked from 'package:survival':
##
##     myeloma
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

Persiapan Data

```r
df = read.csv("C:/Users/admin2/Downloads/WA_Fn-UseC_-Telco-Customer-Churn.csv")

# Data Cleaning: Kalau Churn Yes = 1, sedangkan No = 0
df = df %>%
  mutate(status = ifelse(Churn == "Yes", 1, 0))

df = df %>%
  filter(Contract %in% c("Month-to-month", "One year"))

head(df)
```

```
##   customerID gender SeniorCitizen Partner Dependents tenure PhoneService
## 1 7590-VHVEG Female             0     Yes         No      1           No
## 2 5575-GNVDE   Male             0      No         No     34          Yes
## 3 3668-QPYBK   Male             0      No         No      2          Yes
## 4 7795-CFOCW   Male             0      No         No     45           No
## 5 9237-HQITU Female             0      No         No      2          Yes
## 6 9305-CDSKC Female             0      No         No      8          Yes
##      MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection
## 1 No phone service             DSL             No          Yes               No
## 2               No             DSL            Yes           No              Yes
## 3               No             DSL            Yes          Yes               No
## 4 No phone service             DSL            Yes           No              Yes
## 5               No     Fiber optic             No           No               No
## 6              Yes     Fiber optic             No           No              Yes
##   TechSupport StreamingTV StreamingMovies       Contract PaperlessBilling
## 1          No          No              No Month-to-month              Yes
## 2          No          No              No       One year               No
## 3          No          No              No Month-to-month              Yes
## 4         Yes          No              No       One year               No
## 5          No          No              No Month-to-month              Yes
## 6          No         Yes             Yes Month-to-month              Yes
##               PaymentMethod MonthlyCharges TotalCharges Churn status
## 1          Electronic check          29.85        29.85    No      0
## 2              Mailed check          56.95      1889.50    No      0
## 3              Mailed check          53.85       108.15   Yes      1
## 4 Bank transfer (automatic)          42.30      1840.75    No      0
## 5          Electronic check          70.70       151.65   Yes      1
## 6          Electronic check          99.65       820.50   Yes      1
```

Metode Estimasi Fungsi Survival

Fungsi Empiris

Dengan data utama, tapi menghilangkan data yang tersensor

```
data_tanpa_sensor <- subset(df, status == 1)

objek_surv <- Surv(time = data_tanpa_sensor$tenure, event = data_tanpa_sensor$status)

print(objek_surv)
```

```
##    [1]  2  2  8 28 49 10  1  1 47  1 17  5 34 11  2 15  8 18  9  1  7 12  5 25
##   [25] 55 37 10  3 27 25  1  2  1  8  1  7 20  4 58  2 53 15  7  1  8 13  1  6
##   [49]  1 19  2  4  4  1  1  2 11 15  1  2  1  2  3 13 16  1  1  5 52 18  1  8
##   [73] 10 24  1  4 32  5 38 24 54  2  7  2 68 63  2  4 21 10 27  5  2 25  1  1
##   [97] 11 55 34 69  1  1  3 22  1 61 12 60  1 28  4  7 13  1  1 24  5 15 48  8
##  [121]  1  2 22  5 12 20  2  6  5 40 27 23 34  1 10 13 39 32 35 56 12  1  4  1
##  [145] 12 65  9  9  7 49 43 65 56 10 43  1 20  1  2  3  1  8  2  4  2  4  1 52
##  [169]  2  1  2  3  3 25  1  4  3 33 30 45 10 20 15 22  3 15  1  5 33  2  2  1
##  [193] 46 62 10  6 40  1 58  2 21 16  7 60 50  4  1 55 12  1 62  1 20  3 22  4
##  [217] 59 60 15 10  6 63 50  1 12  9 17  9 10  1  3 13  4  6  3 35  2  1  1  8
##  [241]  7  2  1 44  1 62  1 33 35 34  4  7 27 18  2 32 18 12  8 26  3  8 38 14
##  [265] 32  1 12  5 40  5 10  2 23 23 49  2  2  7 41  2 13 11 17 25  3  4 37 19
##  [289] 39  7 14 11  1  2 13  3  1 12 60  1  6 15 21  1  7 53  1  1 17 11 19  3
##  [313] 24  2  1 16 34  3 16  1 15 11  1  1  7  2 46 68 59  7  8  6 11 15 70 16
##  [337] 18 34 47 39  1  4  1 59  1  2  1 66  3 14 10 18  9 13  4  4 59 10  1  5
##  [361]  1 22 34 41 64  4 18  1  2 15  9 15  1 12  2  5  4 16  1  5  4 37 13 24
##  [385] 61 41 22  2 19  8 15  1 17  1  2 29  2 14  8  5 63  1 53  1 16  1 10  3
##  [409] 66 17  7 12 42  2  1 30 15 13 67 23 35 56  9 17 48 17 34 42 11  1  1  3
##  [433] 50  5 51 46  1 41 29 26  1  3 66 21  1  4  9  1  1  9  1  1  2  3 14 41
##  [457] 13  2 18  1  4 38 12 19 10  1 23 32 32 31 19 57  1 24  1 10  2  1  1  1
##  [481] 38 19 52  1  3  1  2 35 16  1  1 30  5 56 44  1 49 47 31  3  8  2 65  1
##  [505]  1 24 69  7 11  1 58 14 12 25 58 16  1  1  2 23 24 49  5  2  4  5 47 10
##  [529]  1 67  5 67 15  1 26 22 21 47  2 22  7  1 49 43  1  1  1 32 17  4  1  1
##  [553] 37  8  3  1  5 40  1 51  1 33 30  2  2 29 13  7 30  2 24  1  1  7 67 57
##  [577]  7  1 14 28 17 10 34  1  3  2  6 12  1  3 23  1 65 16 66  1 53 13  4 12
##  [601]  1 54 48  2 29 15 17 56 21  1  1  1 35 32 33  1  8  5  2 42 40  1 38  2
##  [625]  8  7 31 18 14 28 15  5  1  1  4 53 59 46 14 31 13 35  2  1 25 16  1 36
##  [649]  1  9  8 45 25 13 46 25 18 60 30  1 39  1 31  3  1  7 61 39 10  1 67  6
##  [673]  1 33  2  7  1 52 22 47 33 18 24  1  4  2  1 19 22  4  4  4  7 19  5  8
##  [697]  1  2  3 13 28 15 10  5  2  1  5  1  4 31  1  1 10 15 17 53 16  9 25  4
##  [721]  9  3 10 32  9 66  3 29  1 41  8 12  1  4  1  7  6 14 25 22 61 70  1  1
##  [745] 47  1 22 48  1 25 20 18  1  9 28 11 12 53 41 20  7  7 22  3 51  5 62 28
##  [769] 10 69 12 27  1 14 32  3 22 16  5 23  1 50  1 46  2  1 32  3  5 49 17 23
##  [793]  6  6  2  4  4  3  8  1  1 30  5  1 13 32 16 29 38 17  1  2  2  1  4  5
##  [817]  9 18 25 17 52  3  1  8 43 22 54  1  1  7 12 54  1 23  3 10 37 31 34 29
##  [841] 12  1 33  4 35 19  1  1 31  4  1 10  2  9  4 38 51  1 41 12  5 10 11  1
##  [865] 16  1  4  1 13  1 21 11  2  2  1  6  4 12  4  4  3 53 68  1 10  1 51 24
##  [889] 18  4  1 26  7  5 42  4  2  8  9  1 43  1  3 70 11 50  5  1 16 15 46  8
##  [913] 31 32  1 20 26  5  1 14 36  9  5  9  2  1  8  2  3 51  2 39 12 26 18  2
##  [937]  4  1  9  2 29  1  9 50  1 54 42  1  1  9 46  1 13 19  1 24  6 20 20 46
##  [961] 11  2  1 16  3 57  5  4  7  1  2  4  1  7  4  1  1  1  1 41 35 10  2 26
```

```
##  [985] 15 31 18  1 24  3  1 43  4 25 57  4 36  1 35  4 30 21  3 41 21  1 11  1
## [1009] 72  3  1 47 18  1 20  6  1 48  3  2  1  1  3  9 49  6  6 16  7 20  4 28
## [1033]  1 11  4 54 28  2  1  1 14  1  4 20 14 30  3 40  2  1 10  1 14 58 22  1
## [1057] 49 68  1  7  9  1 24  2 27 57  1 13  1 63 25 35  6 15  9 37  2 29  5 28
## [1081] 11 55 24  1  6  4  3  1 11  1  5  8 22 29 13 43  3  2 66 62 17 31  2 66
## [1105]  3  1 31 14 35 37  7 24  1 32  1 17 19  2  8  1  1  1  9  3 49  2 21 69
## [1129]  1 39  6 17  1 58  1 19  3  3  3 30 15 10  1 38  1 16  1 12  4 15 41  1
## [1153]  1  7 42  1 12 55 17  1 11  8  1 23  9 15 30  1  3  1  1 12 22  2 27 10
## [1177]  1 22  1 21  6 72 54  3  8  5  1  2  1 33 15 30  2  4  2 25 25 43 10  2
## [1201]  9 27 69 22 56 24  1 36  1  1  1 39  5 24  2  5 11  4  1  1  2  3 44 24
## [1225] 18  1 65 53  1 15  4  9 22 38  1  8 16  1  7  9 37  5  3 31  2 37 38 48
## [1249] 10  8  2  3  4 37  1  5 28 30 36  1  4 17  3  4  4 26  5  4  1  1  8  1
## [1273] 50 26  6  3  3  1 14 61 40 58 18  7  2 19 11 39 15  5  1 10  6 16  2  2
## [1297] 19  1 36 39 21  2  4 35 33 45  7  6  3 11  1 40 47  7  1  7 15 11 67  8
## [1321] 44 24 15 17  1  9  7 36  3 34  1 22 47 24 53  8  2  8 48 43 16  8 20  2
## [1345] 27  5 37 69 40  6 11  6  3  5  8  1 28  1 29 19  6  1  1  6 15 25 10 13
## [1369]  1 37 26 12 10  5 18  3 54 10  1  2 32 36 53  1 12  1  5  6 26  1 64 13
## [1393]  9 10  2 20  9 32  1  1  1 13  7  1  7 54  5  1 33 55  6  8 11 49  1 15
## [1417] 16 44  5  4 30  1  9  1  1 59  3  3  1 13 18  3  1  1 26 11  1  1  4 22
## [1441]  9  7  1 40 29  3  1  3 23 13  1 39  1 14  1  1  1  1  1 42  7  1 10 17
## [1465]  1  8  1 39 58  6 22 70 21 71  1 20  4 32  6 27  1 13  1 19 10 13  1 68
## [1489] 39  1  1 36 15  5 13  5  2 17  1  9  1  1  1  9 20 70  3 20 21 17  2 21
## [1513]  5 24  1 10  1 53  3 46 49 11 45 37 49 14  1 47 13  5  3 26  8  9  1 12
## [1537] 42 31 18 42  4 12 15  7 14  3 16 65 42  6 53  5 68  3 12 43 68 25 66 27
## [1561]  1  3 29 36  1  2 12  1  1 59  3  1  1  2 29 52  2 13 19 47  5 52 42 25
## [1585]  9 16  5  3 31  1  1 10  4 43  4 11 47  1  4 17 43 21 42 46  5  6  1  7
## [1609]  2  1  5 22  1  1  3  1  1 33  1  2  2  3 26  1  5  9 26 54  2  1 30  4
## [1633] 17 60  1 13 38  1 50 65  1  7  2 45  1  1 13 10  4  1 41  3 32 51  6  1
## [1657]  1 31 58 47 44  2  1  1  1  5  1  1  1  1 40  1  3 13 58 42  1 29  1  1
## [1681]  2 10 49  6  2 35 70  2 15 42  9  1 21  2  1 16 14 15  1  2 55  7 13  3
## [1705] 66 38  1 22  1  1 43  1  2  1 12  6 34  1 10 16  7 27  1  9  3  2  2 14
## [1729]  3 57  3  2 12 40 12 54  4  1 51  4 35  1 38 37  3 11  1 19 56  2 14  4
## [1753] 15  1  1 55 18 32 66  7  6 13 11  9  1 54 25 37 14  4  3  1 13 45  1  6
## [1777]  3 12  8  6  3  1  1 41  9 43 33  1 56  3  1  5  1 48 13 24 27 25 22  1
## [1801]  4 18 56 16 30  8 50 41  3  3 40  3 23  1  4  1 12  9  1 67  4
```

```
fit_empiris <- survfit(objek_surv ~ Contract, data = data_tanpa_sensor)

summary(fit_empiris)
```

```
## Call: survfit(formula = objek_surv ~ Contract, data = data_tanpa_sensor)
##
##                 Contract=Month-to-month
##  time n.risk n.event survival  std.err lower 95% CI upper 95% CI
##     1   1655     380 0.770393 0.010338     7.50e-01      0.79092
##     2   1275     121 0.697281 0.011293     6.75e-01      0.71977
##     3   1154      94 0.640483 0.011795     6.18e-01      0.66402
##     4   1060      82 0.590937 0.012086     5.68e-01      0.61510
##     5    978      63 0.552870 0.012222     5.29e-01      0.57735
##     6    915      40 0.528701 0.012270     5.05e-01      0.55331
##     7    875      50 0.498489 0.012290     4.75e-01      0.52317
##     8    825      41 0.473716 0.012274     4.50e-01      0.49839
##     9    784      45 0.446526 0.012220     4.23e-01      0.47113
##    10    739      45 0.419335 0.012130     3.96e-01      0.44380
```

```
##    11    694      30 0.401208 0.012048     3.78e-01      0.42553
##    12    664      33 0.381269 0.011939     3.59e-01      0.40540
##    13    631      37 0.358912 0.011791     3.37e-01      0.38278
##    14    594      23 0.345015 0.011685     3.23e-01      0.36869
##    15    571      37 0.322659 0.011491     3.01e-01      0.34599
##    16    534      28 0.305740 0.011325     2.84e-01      0.32876
##    17    506      24 0.291239 0.011168     2.70e-01      0.31397
##    18    482      22 0.277946 0.011012     2.57e-01      0.30039
##    19    460      16 0.268278 0.010891     2.48e-01      0.29050
##    20    444      18 0.257402 0.010747     2.37e-01      0.27935
##    21    426      15 0.248338 0.010620     2.28e-01      0.27005
##    22    411      23 0.234441 0.010414     2.15e-01      0.25577
##    23    388      12 0.227190 0.010300     2.08e-01      0.24830
##    24    376      23 0.213293 0.010069     1.94e-01      0.23397
##    25    353      18 0.202417 0.009877     1.84e-01      0.22273
##    26    335      15 0.193353 0.009708     1.75e-01      0.21335
##    27    320      13 0.185498 0.009555     1.68e-01      0.20520
##    28    307      10 0.179456 0.009433     1.62e-01      0.19893
##    29    297      15 0.170393 0.009242     1.53e-01      0.18950
##    30    282      15 0.161329 0.009042     1.45e-01      0.18006
##    31    267      14 0.152870 0.008846     1.36e-01      0.17123
##    32    253      18 0.141994 0.008580     1.26e-01      0.15985
##    33    235       8 0.137160 0.008456     1.22e-01      0.15478
##    34    227      11 0.130514 0.008281     1.15e-01      0.14780
##    35    216      12 0.123263 0.008081     1.08e-01      0.14016
##    36    204       9 0.117825 0.007925     1.03e-01      0.13443
##    37    195      14 0.109366 0.007672     9.53e-02      0.12548
##    38    181       9 0.103927 0.007501     9.02e-02      0.11972
##    39    172       9 0.098489 0.007325     8.51e-02      0.11394
##    40    163      11 0.091843 0.007099     7.89e-02      0.10687
##    41    152      10 0.085801 0.006884     7.33e-02      0.10041
##    42    142      12 0.078550 0.006613     6.66e-02      0.09264
##    43    130      11 0.071903 0.006350     6.05e-02      0.08549
##    44    119       4 0.069486 0.006250     5.83e-02      0.08288
##    45    115       3 0.067674 0.006174     5.66e-02      0.08092
##    46    112       7 0.063444 0.005992     5.27e-02      0.07635
##    47    105      13 0.055589 0.005632     4.56e-02      0.06780
##    48     92       3 0.053776 0.005545     4.39e-02      0.06582
##    49     89      10 0.047734 0.005241     3.85e-02      0.05919
##    50     79       8 0.042900 0.004981     3.42e-02      0.05386
##    51     71       6 0.039275 0.004775     3.09e-02      0.04984
##    52     65       5 0.036254 0.004595     2.83e-02      0.04648
##    53     60       6 0.032628 0.004367     2.51e-02      0.04242
##    54     54       5 0.029607 0.004167     2.25e-02      0.03901
##    55     49       6 0.025982 0.003910     1.93e-02      0.03490
##    56     43       3 0.024169 0.003775     1.78e-02      0.03283
##    57     40       5 0.021148 0.003537     1.52e-02      0.02935
##    58     35       4 0.018731 0.003333     1.32e-02      0.02655
##    59     31       4 0.016314 0.003114     1.12e-02      0.02372
##    60     27       3 0.014502 0.002939     9.75e-03      0.02157
##    61     24       3 0.012689 0.002751     8.30e-03      0.01941
##    62     21       1 0.012085 0.002686     7.82e-03      0.01868
##    63     20       2 0.010876 0.002550     6.87e-03      0.01722
##    65     18       4 0.008459 0.002251     5.02e-03      0.01425
```

```
##    66     14       5 0.005438 0.001808     2.83e-03      0.01043
##    67      9       1 0.004834 0.001705     2.42e-03      0.00965
##    68      8       3 0.003021 0.001349     1.26e-03      0.00725
##    69      5       3 0.001208 0.000854     3.02e-04      0.00483
##    70      2       1 0.000604 0.000604     8.52e-05      0.00429
##    71      1       1 0.000000      NaN           NA           NA
##
##                 Contract=One year
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     2    166       2   0.9880 0.00847      0.97149       1.0000
##     4    164       1   0.9819 0.01034      0.96187       1.0000
##     5    163       1   0.9759 0.01190      0.95285       0.9995
##     7    162       1   0.9699 0.01327      0.94422       0.9962
##     8    161       1   0.9639 0.01449      0.93588       0.9927
##     9    160       1   0.9578 0.01560      0.92774       0.9889
##    11    159       1   0.9518 0.01662      0.91978       0.9850
##    12    158       5   0.9217 0.02085      0.88171       0.9635
##    13    153       1   0.9157 0.02157      0.87435       0.9589
##    14    152       1   0.9096 0.02225      0.86705       0.9543
##    17    151       2   0.8976 0.02353      0.85263       0.9449
##    18    149       2   0.8855 0.02471      0.83841       0.9353
##    19    147       3   0.8675 0.02632      0.81739       0.9206
##    21    144       2   0.8554 0.02730      0.80356       0.9106
##    22    142       4   0.8313 0.02906      0.77627       0.8903
##    23    138       1   0.8253 0.02947      0.76951       0.8851
##    25    137       4   0.8012 0.03098      0.74274       0.8643
##    28    133       2   0.7892 0.03166      0.72948       0.8537
##    30    131       1   0.7831 0.03199      0.72288       0.8484
##    31    130       2   0.7711 0.03261      0.70975       0.8377
##    32    128       1   0.7651 0.03291      0.70321       0.8324
##    33    127       5   0.7349 0.03426      0.67077       0.8052
##    34    122       1   0.7289 0.03450      0.66434       0.7998
##    35    121       3   0.7108 0.03519      0.64512       0.7833
##    36    118       1   0.7048 0.03540      0.63874       0.7777
##    37    117       1   0.6988 0.03561      0.63238       0.7722
##    38    116       4   0.6747 0.03636      0.60707       0.7499
##    39    112       4   0.6506 0.03701      0.58197       0.7273
##    40    108       2   0.6386 0.03729      0.56950       0.7160
##    41    106       4   0.6145 0.03778      0.54470       0.6931
##    42    102       2   0.6024 0.03798      0.53238       0.6817
##    43    100       3   0.5843 0.03825      0.51398       0.6643
##    44     97       2   0.5723 0.03840      0.50177       0.6527
##    45     95       3   0.5542 0.03858      0.48354       0.6352
##    46     92       4   0.5301 0.03874      0.45938       0.6117
##    47     88       1   0.5241 0.03876      0.45337       0.6059
##    48     87       5   0.4940 0.03880      0.42349       0.5762
##    49     82       4   0.4699 0.03874      0.39977       0.5523
##    50     78       1   0.4639 0.03871      0.39387       0.5463
##    51     77       2   0.4518 0.03863      0.38210       0.5342
##    52     75       2   0.4398 0.03852      0.37038       0.5221
##    53     73       7   0.3976 0.03798      0.32970       0.4795
##    54     66       7   0.3554 0.03715      0.28958       0.4362
##    55     59       2   0.3434 0.03685      0.27823       0.4238
##    56     57       6   0.3072 0.03581      0.24449       0.3861
```

```
##    57     51       1   0.3012 0.03561      0.23891       0.3797
##    58     50       6   0.2651 0.03426      0.20575       0.3415
##    59     44       3   0.2470 0.03347      0.18937       0.3221
##    60     41       3   0.2289 0.03261      0.17315       0.3026
##    61     38       2   0.2169 0.03199      0.16242       0.2896
##    62     36       4   0.1928 0.03062      0.14120       0.2632
##    63     32       2   0.1807 0.02987      0.13072       0.2498
##    64     30       2   0.1687 0.02906      0.12033       0.2364
##    65     28       3   0.1506 0.02776      0.10494       0.2161
##    66     25       5   0.1205 0.02527      0.07988       0.1817
##    67     20       6   0.0843 0.02157      0.05109       0.1392
##    68     14       4   0.0602 0.01847      0.03303       0.1099
##    69     10       3   0.0422 0.01560      0.02042       0.0871
##    70      7       5   0.0120 0.00847      0.00304       0.0478
##    72      2       2   0.0000     NaN           NA           NA
```

Life Table

```r
# Buat interval tiap 1 tahun
interval_LT = seq(0, max(df$tenure, na.rm=TRUE) + 12, by=12)

# Menghitung Life Table Interval
life_table = df %>%
  mutate(interval = cut(tenure, breaks = interval_LT, right = FALSE)) %>%
  group_by(interval) %>%
  summarise(
    event_churn = sum(status == 1),
    sensor = sum(status == 0)
  ) %>%
  mutate(
    n_risk = nrow(df) - lag(cumsum(event_churn + sensor), default = 0),
    n_risk_adj = n_risk - (sensor / 2),
    q = event_churn / n_risk_adj,
    p = 1 - q,
    Survival = cumprod(p)
  )

print(as.data.frame(life_table))
```

```
##   interval event_churn sensor n_risk n_risk_adj          q         p  Survival
## 1   [0,12)         999   1011   5348     4842.5 0.20629840 0.7937016 0.7937016
## 2  [12,24)         309    655   3338     3010.5 0.10264076 0.8973592 0.7122355
## 3  [24,36)         191    582   2374     2083.0 0.09169467 0.9083053 0.6469273
## 4  [36,48)         143    439   1601     1381.5 0.10351068 0.8964893 0.5799634
## 5  [48,60)         111    444   1019      797.0 0.13927227 0.8607277 0.4991906
## 6  [60,72)          66    379    464      274.5 0.24043716 0.7595628 0.3791666
## 7  [72,84)           2     17     19       10.5 0.19047619 0.8095238 0.3069444
```

Kaplan-Meier

Ingin dibandingkan antara pelanggan dengan kontrak "Month-to-month" dan "One year". Oleh karena itu, kita akan memfilter data untuk hanya menyertakan kedua jenis kontrak tersebut.

```r
# Model Kaplan-Meier
kaplan_meier = survfit(Surv(tenure, status) ~ Contract, data = df)

summary(kaplan_meier)
```

```
## Call: survfit(formula = Surv(tenure, status) ~ Contract, data = df)
##
##                 Contract=Month-to-month
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1   3875     380    0.902 0.00478       0.8926        0.911
##     2   3271     121    0.869 0.00548       0.8579        0.879
##     3   3041      94    0.842 0.00597       0.8301        0.854
##     4   2850      82    0.818 0.00637       0.8051        0.830
##     5   2685      63    0.798 0.00666       0.7854        0.811
##     6   2557      40    0.786 0.00684       0.7725        0.799
##     7   2462      50    0.770 0.00707       0.7561        0.784
##     8   2349      41    0.756 0.00725       0.7424        0.771
##     9   2248      45    0.741 0.00745       0.7268        0.756
##    10   2143      45    0.726 0.00764       0.7109        0.741
##    11   2047      30    0.715 0.00777       0.7000        0.730
##    12   1967      33    0.703 0.00792       0.6877        0.719
##    13   1881      37    0.689 0.00808       0.6736        0.705
##    14   1798      23    0.680 0.00819       0.6646        0.697
##    15   1742      37    0.666 0.00835       0.6498        0.683
##    16   1663      28    0.655 0.00847       0.6384        0.672
##    17   1597      24    0.645 0.00858       0.6283        0.662
##    18   1540      22    0.636 0.00868       0.6189        0.653
##    19   1463      16    0.629 0.00876       0.6118        0.646
##    20   1411      18    0.621 0.00885       0.6037        0.638
##    21   1357      15    0.614 0.00893       0.5966        0.632
##    22   1313      23    0.603 0.00905       0.5857        0.621
##    23   1249      12    0.597 0.00911       0.5797        0.615
##    24   1202      23    0.586 0.00925       0.5681        0.604
##    25   1144      18    0.577 0.00935       0.5587        0.595
##    26   1095      15    0.569 0.00944       0.5506        0.588
##    27   1046      13    0.562 0.00953       0.5434        0.581
##    28   1005      10    0.556 0.00960       0.5376        0.575
##    29    971      15    0.548 0.00970       0.5289        0.567
##    30    923      15    0.539 0.00981       0.5198        0.558
##    31    880      14    0.530 0.00992       0.5110        0.550
##    32    840      18    0.519 0.01006       0.4994        0.539
##    33    794       8    0.513 0.01013       0.4940        0.534
##    34    763      11    0.506 0.01022       0.4864        0.527
##    35    723      12    0.498 0.01034       0.4778        0.518
##    36    680       9    0.491 0.01043       0.4711        0.512
##    37    658      14    0.481 0.01058       0.4604        0.502
##    38    623       9    0.474 0.01067       0.4532        0.495
##    39    596       9    0.467 0.01078       0.4459        0.488
##    40    575      11    0.458 0.01090       0.4368        0.480
##    41    541      10    0.449 0.01102       0.4281        0.471
##    42    512      12    0.439 0.01118       0.4173        0.461
##    43    481      11    0.429 0.01132       0.4070        0.451
##    44    452       4    0.425 0.01138       0.4031        0.448
```

```
##    45    432       3    0.422 0.01143       0.4001        0.445
##    46    410       7    0.415 0.01155       0.3926        0.438
##    47    384      13    0.401 0.01180       0.3782        0.424
##    48    357       3    0.397 0.01186       0.3747        0.421
##    49    342      10    0.386 0.01207       0.3627        0.410
##    50    313       8    0.376 0.01225       0.3525        0.401
##    51    294       6    0.368 0.01240       0.3446        0.393
##    52    276       5    0.361 0.01252       0.3377        0.387
##    53    252       6    0.353 0.01271       0.3288        0.379
##    54    236       5    0.345 0.01287       0.3210        0.372
##    55    219       6    0.336 0.01309       0.3112        0.363
##    56    195       3    0.331 0.01322       0.3058        0.358
##    57    176       5    0.321 0.01350       0.2960        0.349
##    58    156       4    0.313 0.01377       0.2873        0.341
##    59    143       4    0.304 0.01406       0.2780        0.333
##    60    127       3    0.297 0.01433       0.2704        0.327
##    61    108       3    0.289 0.01470       0.2615        0.319
##    62     94       1    0.286 0.01486       0.2581        0.316
##    63     85       2    0.279 0.01525       0.2508        0.311
##    65     62       4    0.261 0.01672       0.2303        0.296
##    66     46       5    0.233 0.01912       0.1981        0.273
##    67     35       1    0.226 0.01970       0.1906        0.268
##    68     27       3    0.201 0.02221       0.1618        0.250
##    69     17       3    0.165 0.02607       0.1215        0.225
##    70     11       1    0.150 0.02771       0.1049        0.216
##    71      7       1    0.129 0.03098       0.0805        0.207
##
##                  Contract=One year
##  time n.risk n.event survival  std.err lower 95% CI upper 95% CI
##     2   1465       2    0.999 0.000965        0.997        1.000
##     4   1452       1    0.998 0.001184        0.996        1.000
##     5   1445       1    0.997 0.001370        0.995        1.000
##     7   1434       1    0.997 0.001535        0.994        1.000
##     8   1420       1    0.996 0.001687        0.993        0.999
##     9   1411       1    0.995 0.001828        0.992        0.999
##    11   1385       1    0.994 0.001962        0.991        0.998
##    12   1371       5    0.991 0.002539        0.986        0.996
##    13   1349       1    0.990 0.002641        0.985        0.995
##    14   1334       1    0.989 0.002741        0.984        0.995
##    17   1299       2    0.988 0.002941        0.982        0.994
##    18   1273       2    0.986 0.003134        0.980        0.992
##    19   1259       3    0.984 0.003408        0.977        0.991
##    21   1233       2    0.982 0.003584        0.975        0.989
##    22   1220       4    0.979 0.003918        0.971        0.987
##    23   1198       1    0.978 0.003999        0.970        0.986
##    25   1152       4    0.975 0.004331        0.966        0.983
##    28   1085       2    0.973 0.004505        0.964        0.982
##    30   1044       1    0.972 0.004596        0.963        0.981
##    31   1025       2    0.970 0.004779        0.961        0.980
##    32   1009       1    0.969 0.004870        0.960        0.979
##    33    990       5    0.964 0.005315        0.954        0.975
##    34    967       1    0.963 0.005402        0.953        0.974
##    35    949       3    0.960 0.005664        0.949        0.972
##    36    921       1    0.959 0.005753        0.948        0.971
```

```
##    37    902       1    0.958 0.005844        0.947        0.970
##    38    883       4    0.954 0.006208        0.942        0.966
##    39    862       4    0.949 0.006561        0.937        0.962
##    40    840       2    0.947 0.006738        0.934        0.961
##    41    819       4    0.943 0.007091        0.929        0.957
##    42    791       2    0.940 0.007270        0.926        0.955
##    43    770       3    0.937 0.007543        0.922        0.951
##    44    747       2    0.934 0.007729        0.919        0.949
##    45    724       3    0.930 0.008013        0.915        0.946
##    46    704       4    0.925 0.008392        0.909        0.941
##    47    681       1    0.924 0.008489        0.907        0.940
##    48    662       5    0.917 0.008980        0.899        0.934
##    49    634       4    0.911 0.009377        0.893        0.929
##    50    612       1    0.909 0.009479        0.891        0.928
##    51    581       2    0.906 0.009701        0.887        0.925
##    52    546       2    0.903 0.009946        0.884        0.923
##    53    519       7    0.891 0.010824        0.870        0.912
##    54    483       7    0.878 0.011715        0.855        0.901
##    55    460       2    0.874 0.011971        0.851        0.898
##    56    438       6    0.862 0.012766        0.837        0.887
##    57    402       1    0.860 0.012913        0.835        0.885
##    58    385       6    0.846 0.013822        0.820        0.874
##    59    357       3    0.839 0.014303        0.812        0.868
##    60    337       3    0.832 0.014812        0.803        0.861
##    61    313       2    0.827 0.015186        0.797        0.857
##    62    283       4    0.815 0.016056        0.784        0.847
##    63    257       2    0.808 0.016545        0.777        0.842
##    64    236       2    0.802 0.017100        0.769        0.836
##    65    207       3    0.790 0.018120        0.755        0.826
##    66    189       5    0.769 0.019905        0.731        0.809
##    67    154       6    0.739 0.022578        0.696        0.785
##    68    120       4    0.715 0.024961        0.667        0.765
##    69     95       3    0.692 0.027362        0.640        0.748
##    70     72       5    0.644 0.032834        0.583        0.712
##    72     17       2    0.568 0.058060        0.465        0.694
```

Uji Log-Rank

H0: Tidak ada perbedaan kurva survival antar kelompok H1: Ada perbedaan kurva survival antar kelompok

```r
test_logrank = survdiff(Surv(tenure, status) ~ Contract, data = df)

print(test_logrank)
```

```
## Call:
## survdiff(formula = Surv(tenure, status) ~ Contract, data = df)
##
##                             N Observed Expected (O-E)^2/E (O-E)^2/V
## Contract=Month-to-month 3875     1655     1047       353       926
## Contract=One year       1473      166      774       478       926
##
##  Chisq= 926  on 1 degrees of freedom, p= <2e-16
```

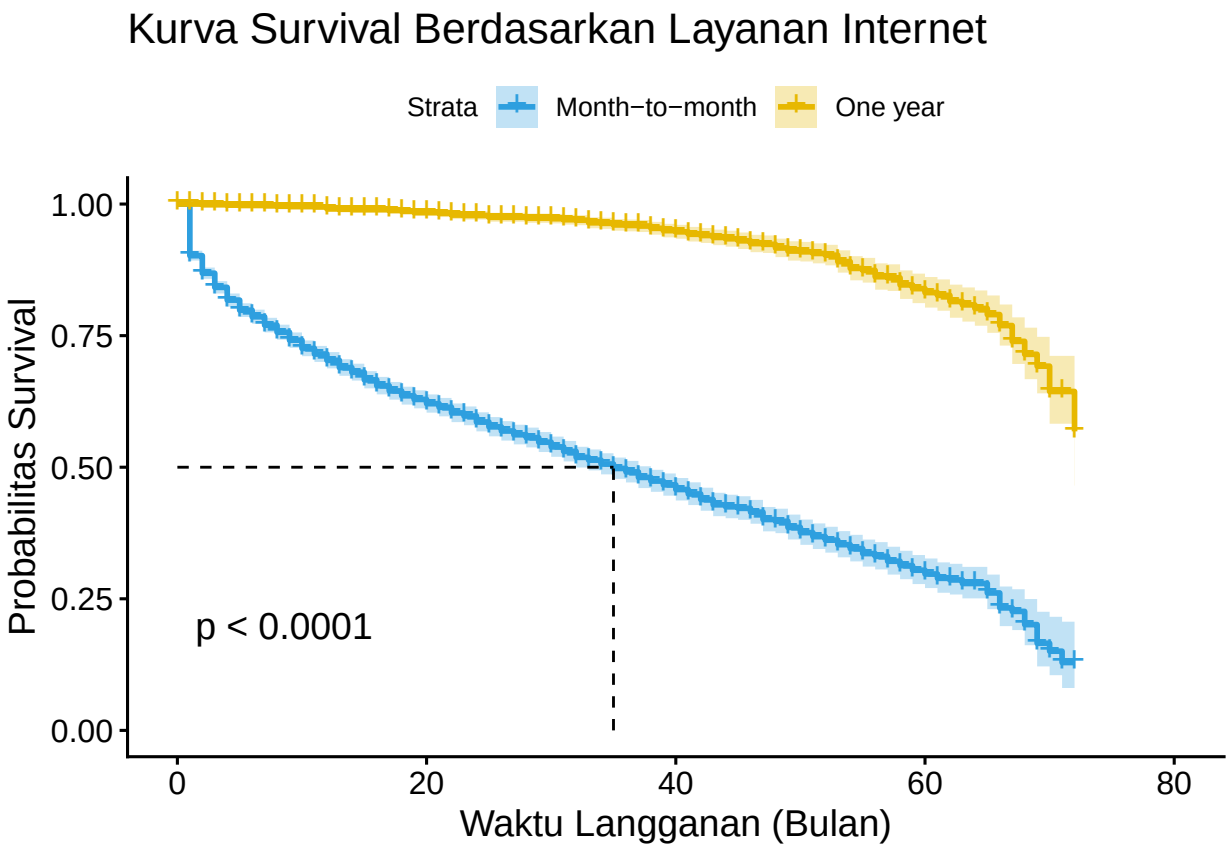
Karena p-value < 0.05, maka tolak H0. Artinya, terdapat perbedaan daya tahan yang signifikan secara statistik antara pelanggan dengan kontrak "Month-to-month" dan "One year".

Visualisasi Kurva

Visualisasi kurva kaplan-meier dengan tambahan nilai p-value dari uji log-rank

```r
ggsurvplot(kaplan_meier,
           data = df,
           pval = TRUE,
           conf.int = TRUE,
           surv.median.line = "hv",
           legend.labs = c("Month-to-month", "One year"),
           palette = c("#2E9FDF", "#E7B800"),
           title = "Kurva Survival Berdasarkan Layanan Internet",
           xlab = "Waktu Langganan (Bulan)",
           ylab = "Probabilitas Survival")
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## i The deprecated feature was likely used in the ggpubr package.
##   Please report the issue at <https://github.com/kassambara/ggpubr/issues>.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```


Kurva Survival Berdasarkan Layanan Internet
Strata
Month−to−month
One year
Probabilitas Survival
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
20
40
60
80
Waktu Langganan (Bulan)